CSTE4103 — Cloud Computing Lab

Lab 4: Clone, Export, and Import Virtual Machines

Linked and full clones, password recovery, and moving a VM with OVF

Week of 10 Nov • Cybersecurity Engineering — 4th Year

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LAB 4 • OVERVIEW

Objectives and Deliverable

By the end you can

- Create a linked clone and a full clone from a snapshot
- Explain the difference in time, disk space, and independence
- Fix what a clone inherits: name, password, machine ID
- Recover a lost password using recovery mode
- Export a VM to OVF and import it as a new VM
- Say what OVF keeps and what it loses

Your deliverable

- One short PDF report
- 4 screenshots: linked clone details (Clone of), full clone details, exported files, imported VM running
- A table comparing linked and full clones with your own numbers
- Answers to the 5 review questions

Cloning: Linked or Full

Install once, then copy. A clone is a new VM made from an existing one, so ten identical lab machines cost minutes, not hours.

	Linked clone	Full clone
What it is	Points to the parent's disk and stores only its own changes	A complete, separate copy of the disk
Disk space	Very small	About the size of the parent's used data
Speed	Seconds	Longer: the disk is copied
Independence	Needs the parent's files to run	Fully independent
Best for	Quick labs and tests	Backups, moving, long-term use

⚠ Security: A clone inherits everything: the password, the SSH keys, the hostname, and the machine ID. Change them before the clone is used by someone else.

Export and Import: OVF and OVA

OVF (Open Virtualization Format) is an open standard that describes a VM so another hypervisor can rebuild it. Use it to move, share, or archive a VM.

File	What it holds
.ovf	XML description: CPU, memory, disks, network
.vmdk	The virtual disk with the OS and data
.mf	Manifest: SHA checksums to detect damaged files
.iso	A CD/DVD image, only if the VM had one attached

OVF: several files

A folder of files. Easy to inspect and edit the .ovf description.

OVA: one file

The same content packed into one .ova file. Easier to copy and send.

⚠ Security: An exported VM carries its whole disk, users, and data. Share it only with people you trust, and keep it off public links.

LAB 4 ● 2 — BEFORE YOU START

Instructions and Warnings

Before cloning or exporting

- Shut the VM down properly (sudo poweroff). Never close the window while it runs.
- Check State: Powered off, not Suspended.
- Clone from a clean snapshot so every copy starts the same.
- Keep at least 6 GB free on the disk.

After cloning

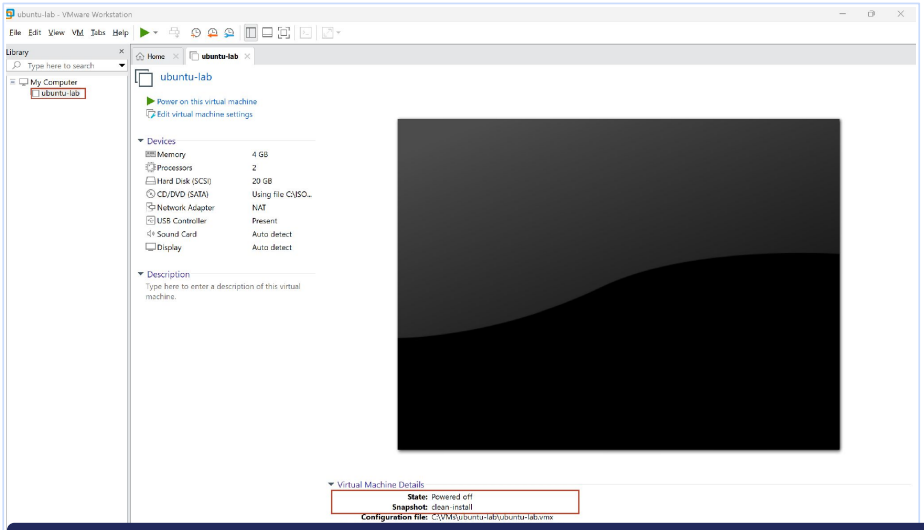
- Change the hostname so two machines do not share a name.
- Do not delete the parent while a linked clone depends on it.
- Change the password before handing the clone to someone else.
- Run the clone and the parent on one network only after renaming.

Reading the screenshots

Red boxes mark what to look at. The numbers beside each picture explain it. Reproduce every step on your own PC and use your own screenshots in the report.

LAB 4 ● 3 — CLONE

Step 1: Start From a Powered-Off VM with a Snapshot



1 **Select the parent VM**

ubuntu-lab in the Library.

2 **State: Powered off**

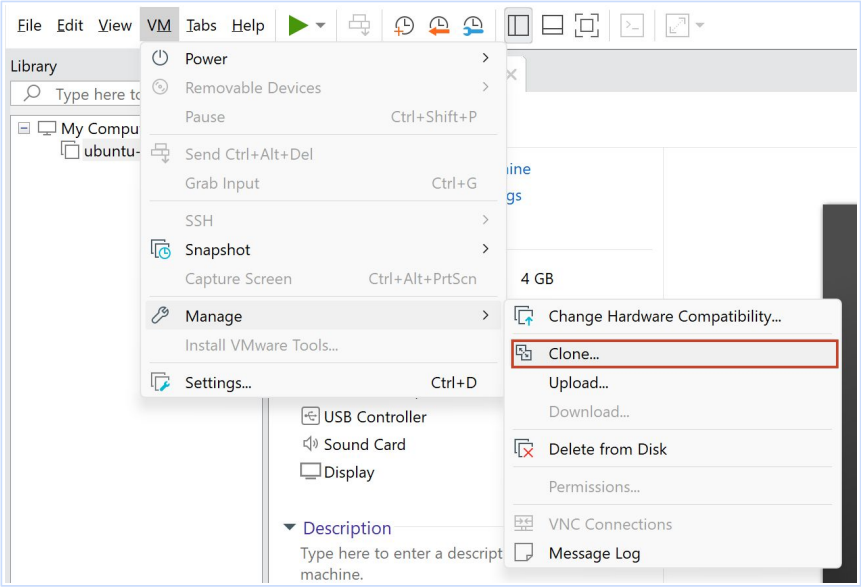
Clone is enabled only for a stopped VM.

3 **Snapshot: clean-install**

A saved clean state, the source of every clone.

VM specs: 4 GB RAM, 2 CPUs, 20 GB disk, NAT network.

Step 2: Open the Clone Wizard

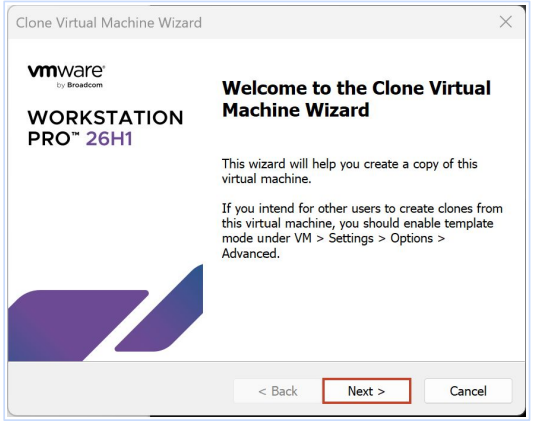


1. **Open the VM menu**
   Choose Manage.

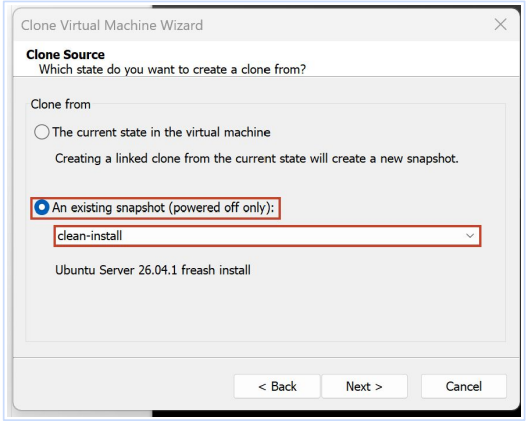
2. **Click Clone…**
   The Clone Virtual Machine Wizard opens.

3. **Grayed out?**
   The VM is running or suspended. Shut it down first.

Steps 3–4: Welcome and Clone Source



Welcome page: click Next



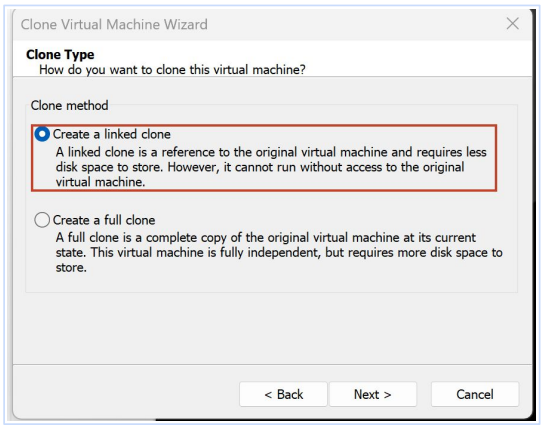
Clone Source

1. **Template mode**
   To let other users clone this VM, enable it in VM > Settings > Options > Advanced.

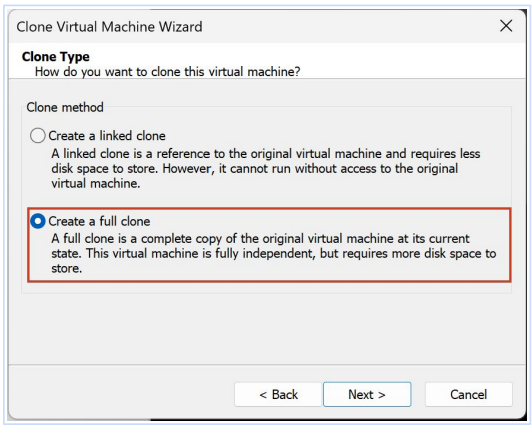
2. **Choose the snapshot**
   Select An existing snapshot, then clean-install. Every clone starts from the same known state.

3. **Current state**
   The first option would include any later changes, and a linked clone would create a new snapshot.

Step 5: Clone Type, the Only Difference



Linked clone



Full clone

1 Linked clone

A reference to the original. Needs less disk space, but cannot run without the original VM.

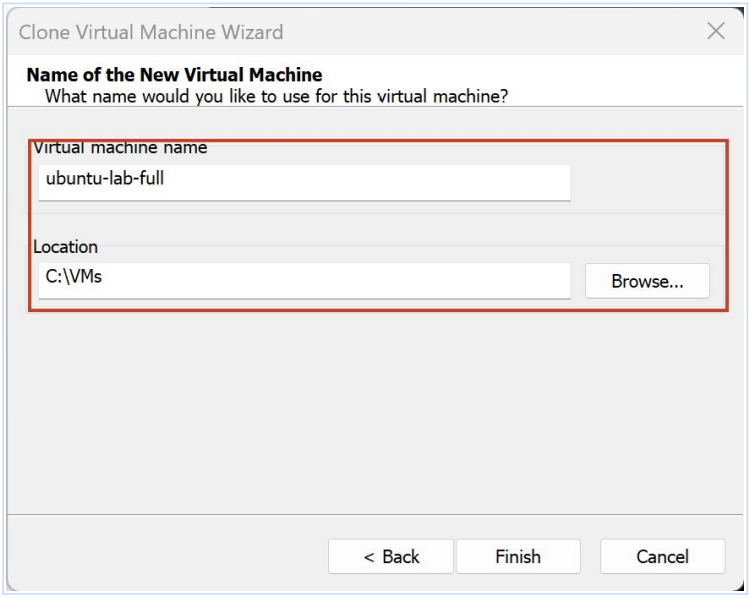
2 Full clone

A complete copy. Fully independent, but needs more disk space and more time.

3 Same steps for both

Everything else in the wizard is identical. Do the lab once for each type.

Step 6: Name and Location



1 Use a clear name

The default is Clone of ubuntu-lab. Use ubuntu-lab-linked or ubuntu-lab-full instead.

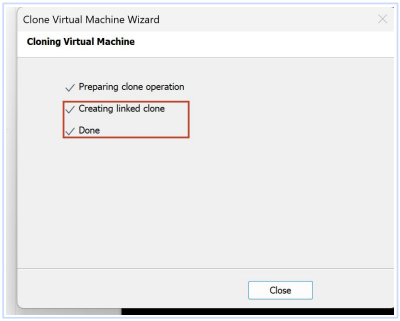
2 Keep VMs in one folder

Browse… to C:\VMs. The default path sits under your Windows profile.

3 Click Finish

Note the clock time so you can measure how long it took.

Linked Clone: Done in Seconds



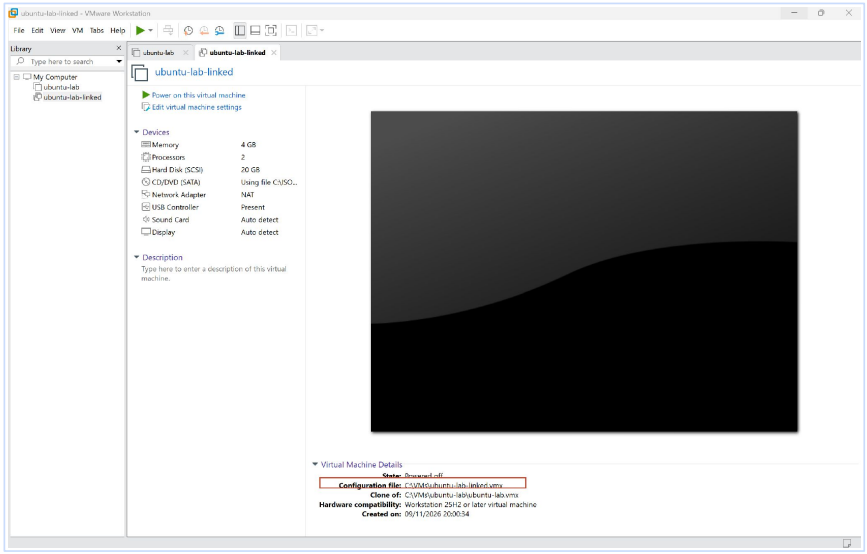
Creating linked clone ✓ Done

1 **No disk copy**

Only a small file of changes is created.

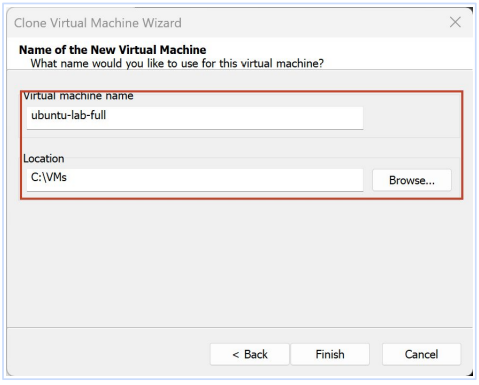
2 **Clone of**

Points to the parent's .vmx. This line proves the VM is linked.

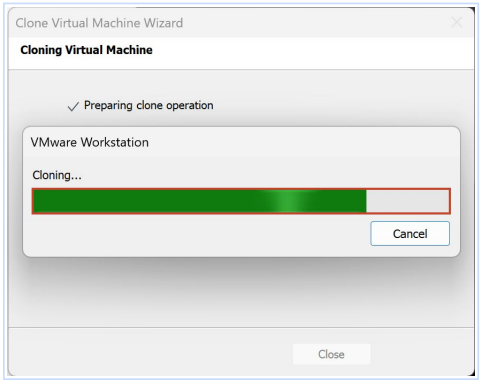


The new VM: note the Clone of line

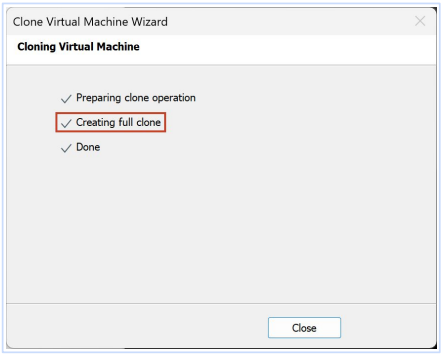
Full Clone: The Disk Is Really Copied



Name: ubuntu-lab-full



A progress bar appears



Creating full clone ✓ Done

1 **Progress bar**

Never shown for a linked clone: here the disk is really copied.

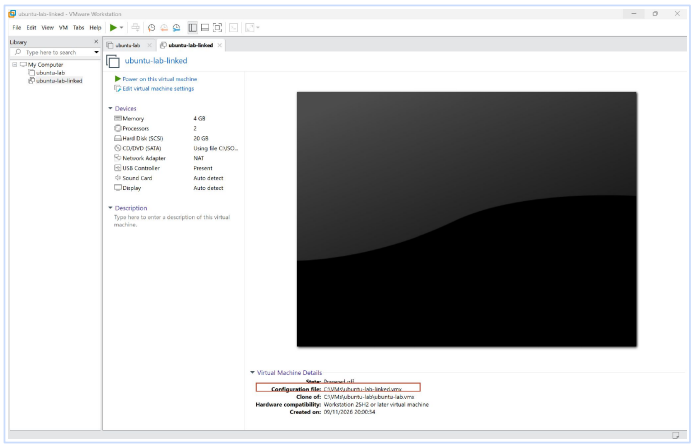
2 **Our measurement**

About 1 minute (21:46 to 21:47) on this PC. A slower disk takes longer.

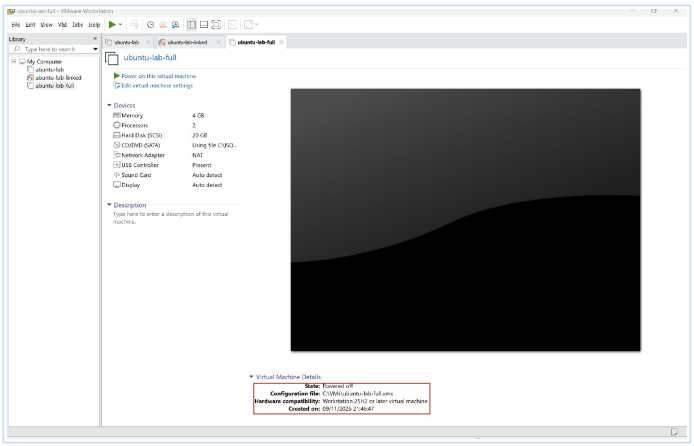
3 **In the lab room**

Thirty students copying at once share one disk and slow each other down.

Compare the Two Clones



Linked: has Clone of



Full: no Clone of line

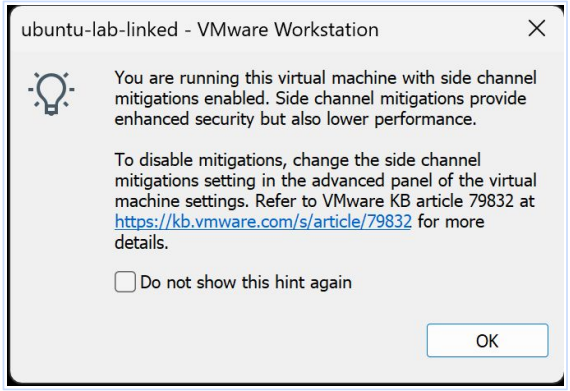
1 Linked

Configuration file plus Clone of C:\VMs\ubuntu-lab\ubuntu-lab.vmx.

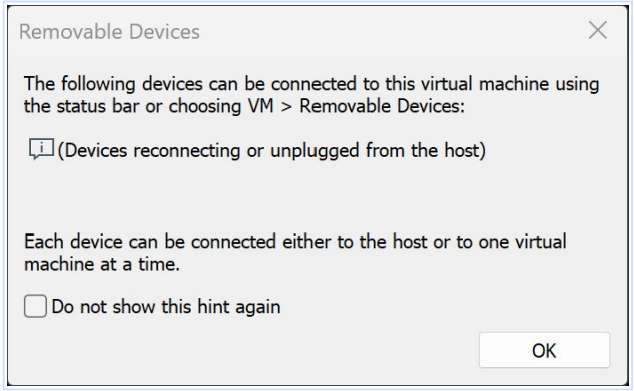
2 Full

Only its own configuration file. It can be moved or kept without the parent.

Step 7: Power On and Two Normal Messages



Side channel mitigations



Removable devices

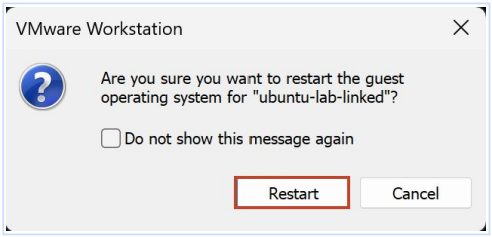
1 Side channel mitigations

Protection against attacks such as Spectre and Meltdown, at a small cost in speed. Click OK.

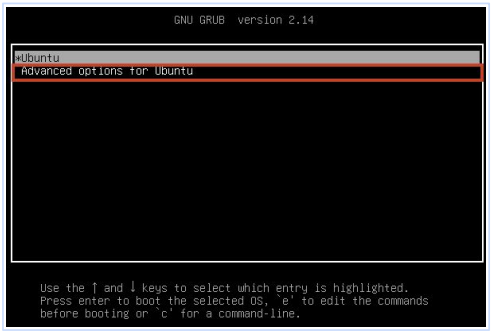
2 Removable devices

A USB or optical device is connected to the host or to one VM, never both. Click OK.

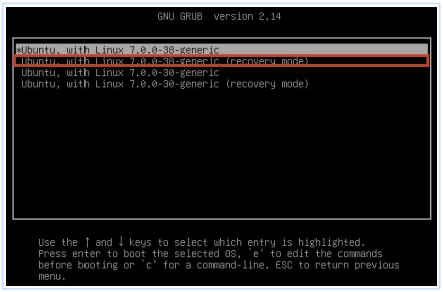
Forgot the Password? Boot Into Recovery



VM > Power > Restart Guest



GRUB: Advanced options



Choose (recovery mode)

1 Restart the guest

Then click inside the screen and press Esc repeatedly until the GRUB menu appears.

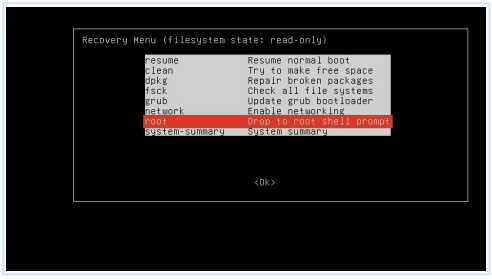
2 Advanced options

Pick Advanced options for Ubuntu with the arrow keys and press Enter.

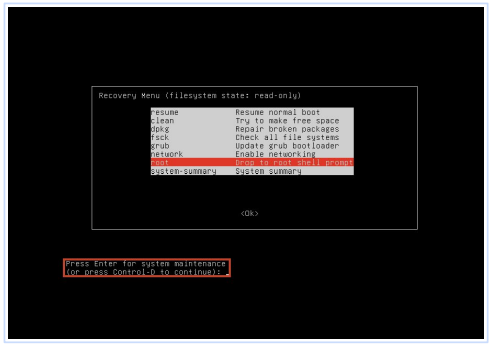
3 Recovery mode

Choose the line that ends with (recovery mode).

Drop to a Root Shell



Recovery Menu: root



Press Enter for maintenance



Two commands

1 Select root

Drop to root shell prompt. The file system starts read-only.

2 Press Enter

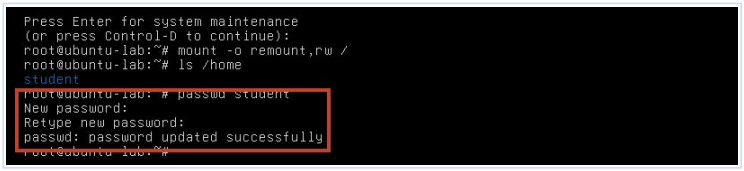
Opens the maintenance shell as root without asking for a password.

3 Two commands

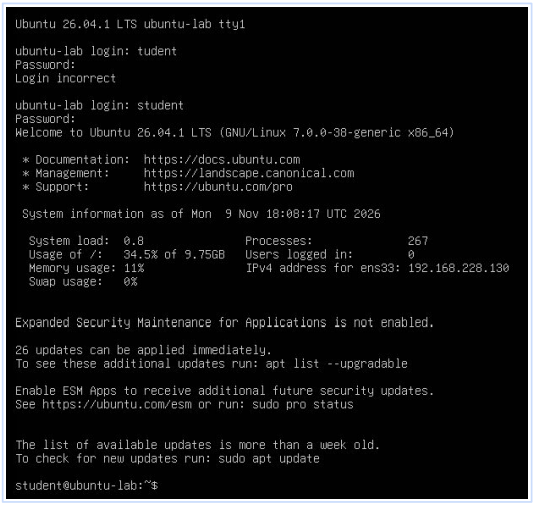
mount -o remount,rw / makes the disk writable. ls /home shows the user name: student.

LAB 4 ● 5 — PASSWORD RECOVERY

Set a New Password and Log In



passwd student



1 **passwd student**

Type the new password twice. Nothing is shown while typing. Success: password updated successfully.

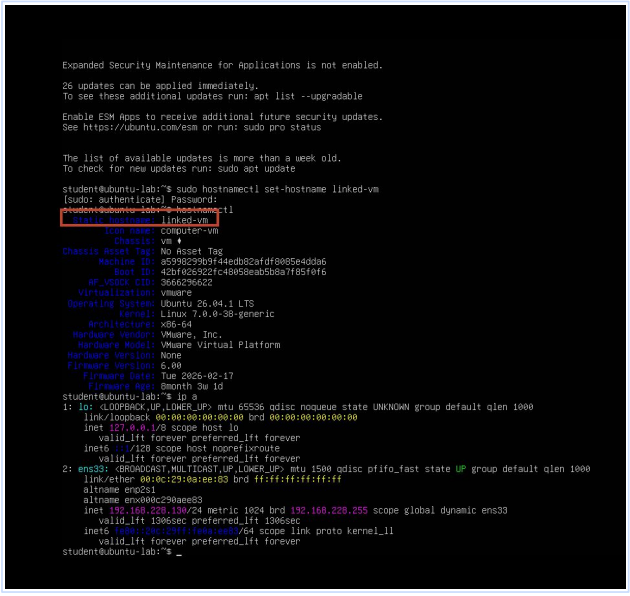
2 **reboot**

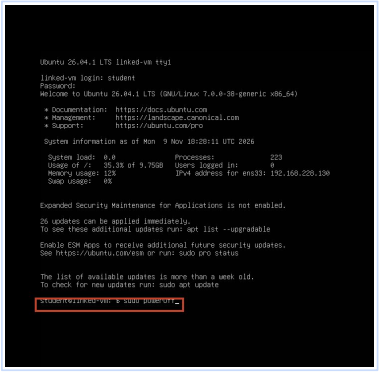
Log in with student and the new password. A typo gives Login incorrect: try again.

⚠ Security: Anyone with access to the VM can do this. That is why real servers use disk encryption and a GRUB password.

LAB 4 ● 4 — RUN THE CLONE

Fix the Hostname and Check the Clone





New name after login

1 **hostnamectl set-hostname**

No message means success. The prompt changes after the next login.

2 **hostnamectl and ip a**

Check the name, Virtualization: vmware, the Machine ID, and the MAC and IP address.

3 **sudo poweroff**

Always shut down properly before the next step.

⚠ Observe: VMware gave the clone a new MAC address, but the Machine ID is copied with the disk. Two machines with one ID confuse DHCP and monitoring. Fix: sudo rm /etc/machine-id, then sudo systemd-machine-id-setup.

LAB 4 ● 4 — RUN THE CLONE

Our Results: Linked vs Full

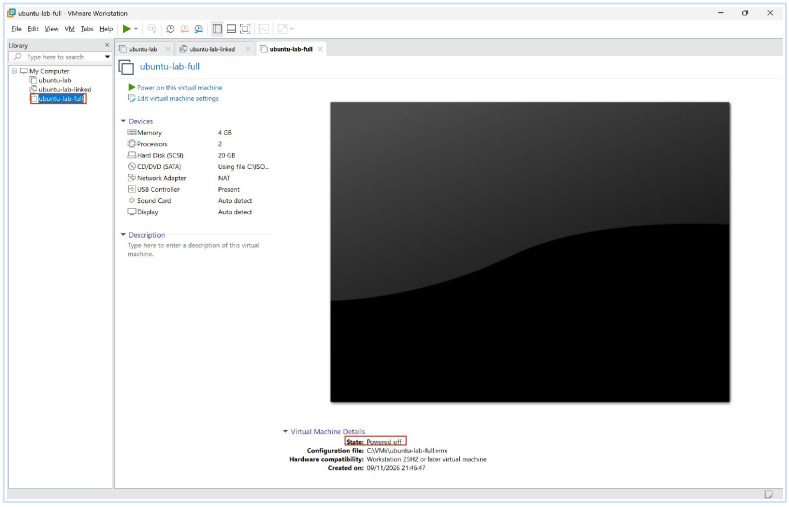
Measure	Linked clone	Full clone
Time to create	A few seconds	About 1 minute
Disk used by its own files	≈ 200 MB	≈ 4.9 GB
Clone of line	Yes	No
Works without the parent	No	Yes
Password and Machine ID	Inherited from the parent	Inherited from the parent

The full clone is about 25 times larger in files. The linked clone saves space and time, but it depends on the parent's snapshot. Choose by the job: quick labs → linked; backups and moving → full.

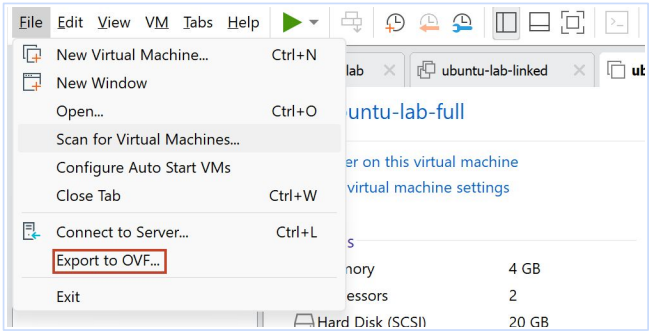
Measured on one lab PC. Your numbers will differ: record your own in the report.

LAB 4 ● 6 — EXPORT

Step 8: Select the VM and Choose Export to OVF



ubuntu-lab-full is selected and Powered off



File > Export to OVF...

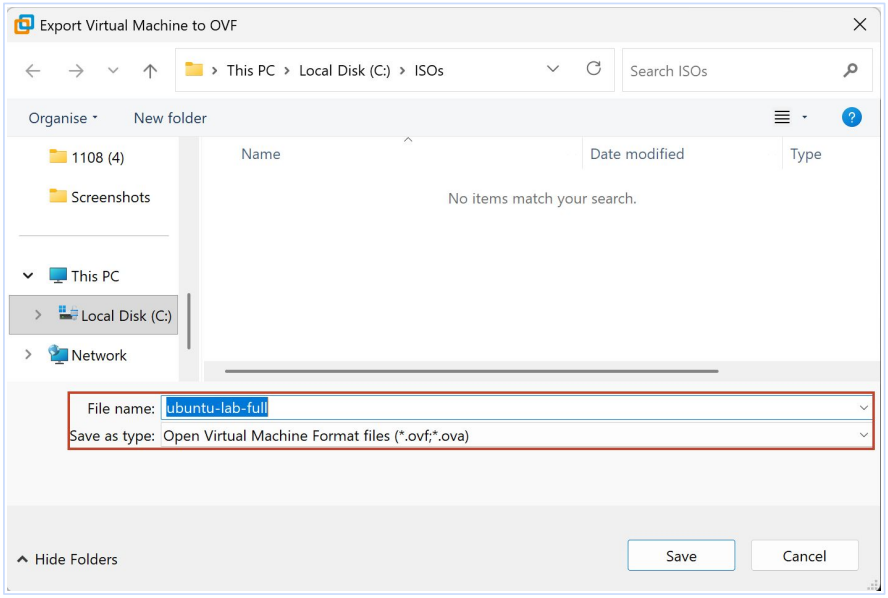
1 **Select the VM**

Click ubuntu-lab-full. State must be Powered off.

2 **File > Export to OVF...**

The command is available only for a stopped VM.

Step 9: Choose the Folder, Name, and Format



1 Folder

The dialog opened in C:\ISOs. Use New folder and create C:\VMs\Exports.

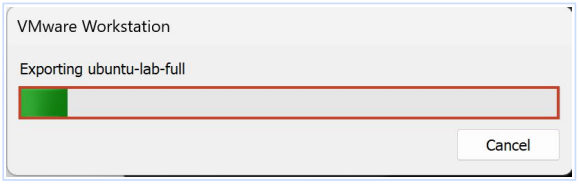
2 File name decides the format

ubuntu-lab-full → an OVF folder of 4 files.
ubuntu-lab-full.ova → one single file.

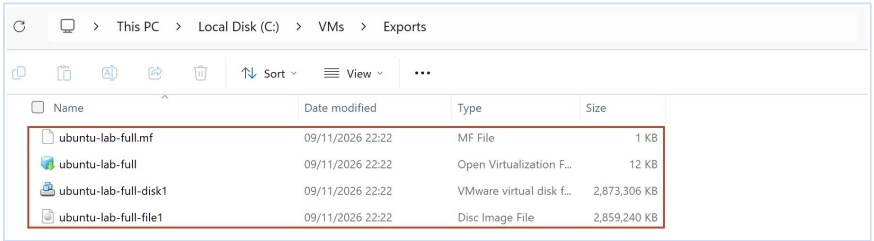
3 Click Save

The export starts.

Step 10: Progress and the Exported Files



Exporting… (about 4 minutes here)

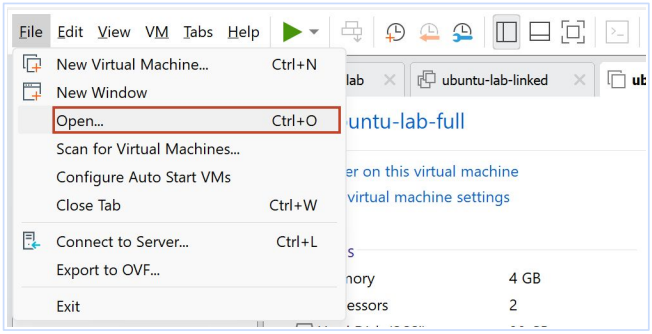


C:\VMs\Exports

File	Size	Job
ubuntu-lab-full.ovf	12 KB	Hardware description (XML)
…-disk1.vmdk	≈ 2.7 GB	The virtual disk
…-file1.iso	≈ 2.7 GB	The install DVD that was still attached
ubuntu-lab-full.mf	1 KB	SHA checksums

⚠ Tip: Disconnect the CD/DVD before exporting (Settings > CD/DVD) and the 2.7 GB .iso is not copied.

Step 11: Import Is File > Open



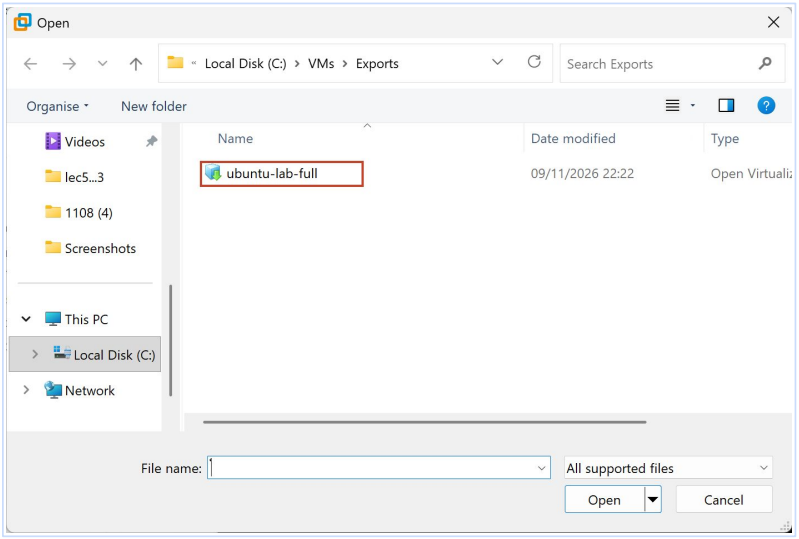
File > Open... (Ctrl+O)

1 **No Import command**

Workstation imports an OVF or OVA through Open.

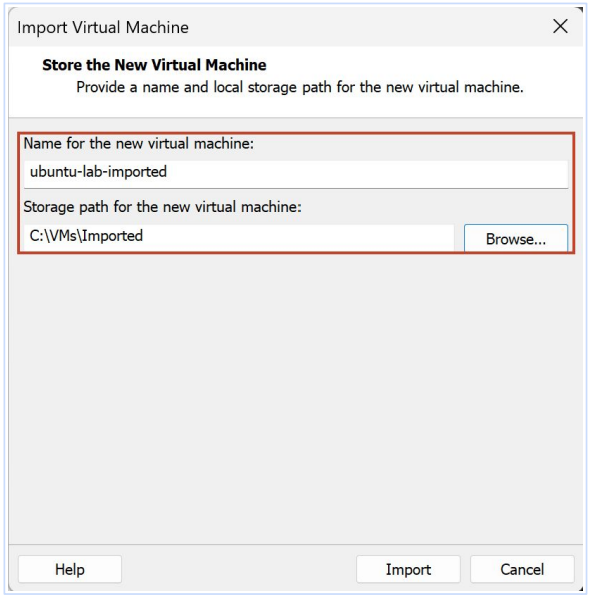
2 **Select the .ovf**

Type: Open Virtualization Format. Then click Open.



Pick the .ovf file

Step 12: Name, Store, and Import





Importing... (about 1 minute)

1 **Name**

ubuntu-lab-imported. The imported VM is a new, separate machine.

2 **Storage path**

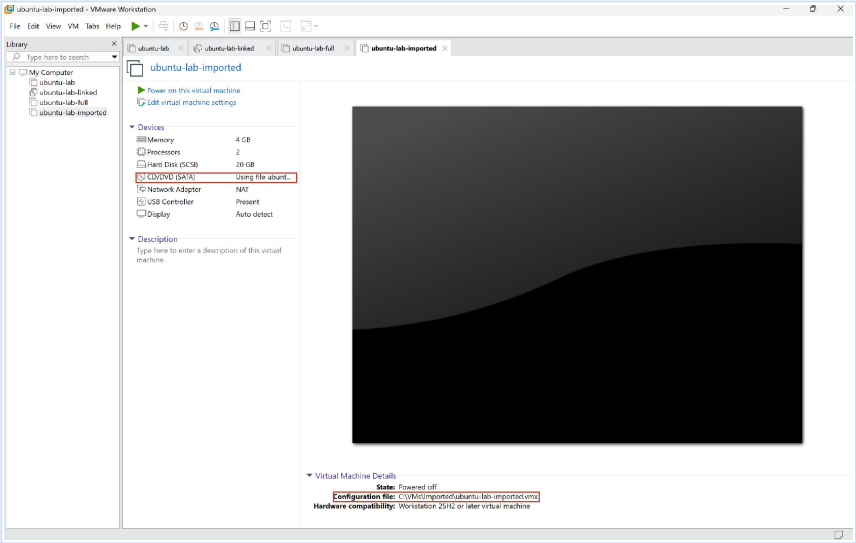
C:\VMs\Imported. VMware copies the disk here.

3 **Click Import**

Wait for the progress bar to finish.

LAB 4 ● 7 — IMPORT

What Changed After the Import



1. **Same hardware**
   4 GB RAM, 2 CPUs, 20 GB disk, NAT.

2. **Sound Card is gone**
   OVF does not carry every device.

3. **CD/DVD points to a copy**
   The ISO was copied into the Imported folder.

4. **No snapshots, no Clone of**
   Only the current state is exported.

Power it on and test it. The password is the one inside the exported disk, not the one you reset on the linked clone.

LAB 4 ● 8 — TROUBLESHOOTING

Common Problems and Fixes

Problem	Cause	Fix
Clone… is grayed out	VM is running or suspended	Shut it down (Powered off)
Keyboard does not type in the VM	Input is still with Windows	Click inside the screen or press Ctrl+G
Login incorrect	A typo, or the wrong password (nothing shows while typing)	Retype it, or reset it in recovery mode
Linked clone will not start	The parent's files were moved or deleted	Put the parent back in place
Export file is huge	The install ISO is still attached	Disconnect the CD/DVD, then export again
Cannot export	VM is running or suspended	Power it off first

REVIEW

Review Questions

1. Why is a linked clone faster and smaller than a full clone? What is its main risk?
2. You must give 30 students identical VMs on lab PCs with little disk space. Which clone type do you choose, and why?
3. Name the four files produced by an OVF export and the job of each.
4. Name two things the imported VM lost compared with the original.
5. A clone keeps the parent's password and machine ID. Why is that a security problem, and how do you fix it?

Deliverable: one PDF with your 4 screenshots, your comparison table, and these answers.